

Proof of Lemma 1. First, we study the relationship between H_{SS} and its population version to be defined later. For a given sample $X = x$, we have the following Hessian matrix for the negative log likelihood of the cGGM:

$$H = \nabla_{\Lambda, \Theta}^2 l_2(\Lambda, \Theta) = \begin{bmatrix} \Lambda^{-1} \otimes (\Lambda^{-1} + 2\Lambda^{-1}\Theta^T S_{xx} \Theta \Lambda^{-1}) & -2\Lambda^{-1} \otimes \Lambda^{-1}\Theta^T S_{xx} \\ -2\Lambda^{-1} \otimes S_{xx} \Theta \Lambda^{-1} & 2\Lambda^{-1} \otimes S_{xx} \end{bmatrix} = \begin{bmatrix} H_1 & H_2 \\ H_3 & H_4 \end{bmatrix},$$

where $H_1 = \Lambda^{-1} \otimes \Lambda^{-1} + \Lambda^{-1} \otimes A_1 S_{xx} B_1$, $H_l = \Lambda^{-1} \otimes A_l S_{xx} B_l$ for $l = 2, 3, 4$, and A_l and B_l are some matrices depending on Λ and Θ . Denote $P = \begin{bmatrix} P_1 & P_2 \\ P_3 & P_4 \end{bmatrix}$ as the permutation matrix such that

$$\begin{bmatrix} P_1 & P_2 \\ P_3 & P_4 \end{bmatrix} \begin{bmatrix} H_1 & H_2 \\ H_3 & H_4 \end{bmatrix} \begin{bmatrix} P_1^T & P_3^T \\ P_2^T & P_4^T \end{bmatrix} = \begin{bmatrix} H_{SS} & H_{S\bar{S}} \\ H_{\bar{S}S} & H_{\bar{S}\bar{S}} \end{bmatrix}, \quad (1)$$

where $H_{SS} = P_1 H_1 P_1^T + P_2 H_3 P_1^T + P_1 H_2 P_2^T + P_2 H_4 P_2^T$, and $H_{\bar{S}S} = P_3 H_1 P_1^T + P_4 H_3 P_1^T + P_3 H_2 P_2^T + P_4 H_4 P_2^T$. Let $H_1 = H_{1,0} + H_{1,1}$ where $H_{1,0} = \Lambda^{-1} \otimes \Lambda^{-1}$. Then,

$$H_{SS} = \begin{bmatrix} P_1 & P_2 \end{bmatrix} \begin{bmatrix} H_1 & H_2 \\ H_3 & H_4 \end{bmatrix} \begin{bmatrix} P_1^T \\ P_2^T \end{bmatrix} = P_1 H_{1,0} P_1^T + P_{1,2} \begin{bmatrix} H_{1,1} & H_2 \\ H_3 & H_4 \end{bmatrix} P_{1,2}^T \quad (2)$$

where $P_{1,2} = \begin{bmatrix} P_1 & P_2 \end{bmatrix}$. Note that the first part of H_{SS} is positive definite because $H_{1,0}$ is positive definite. In the second part, we can rewrite

$$\begin{aligned} \begin{bmatrix} H_{1,1} & H_2 \\ H_3 & H_4 \end{bmatrix} &= \begin{bmatrix} 2\Lambda^{-1} \otimes \Lambda^{-1}\Theta^T S_{xx} \Theta \Lambda^{-1} & -2\Lambda^{-1} \otimes \Lambda^{-1}\Theta^T S_{xx} \\ -2\Lambda^{-1} \otimes S_{xx} \Theta \Lambda^{-1} & 2\Lambda^{-1} \otimes S_{xx} \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2}\Lambda^{-1/2} \otimes \Lambda^{-1}\Theta^T S_{xx}^{1/2} \\ -\sqrt{2}\Lambda^{-1/2} \otimes S_{xx}^{1/2} \end{bmatrix} \begin{bmatrix} \sqrt{2}\Lambda^{-1/2} \otimes S_{xx}^{1/2} \Theta \Lambda^{-1} & -\sqrt{2}\Lambda^{-1/2} \otimes S_{xx}^{1/2} \end{bmatrix} \\ &= UU^T. \end{aligned}$$

Therefore, $P_{1,2} \begin{bmatrix} H_{1,1} & H_2 \\ H_3 & H_4 \end{bmatrix} P_{1,2}^T = P_{1,2} U (P_{1,2} U)^T$ is positive semi-definite. Thus,

there exists some constant C_P such that $|||H_{SS}^{-1}|||_2 = |||(P_1 H_{1,0} P_1^T + P_{1,2} U (P_{1,2} U)^T)^{-1}|||_2 \leq |||(P_1 H_{1,0} P_1^T)^{-1}|||_2 \leq C_P < \infty$.

The population version of the Hessian for the negative log likelihood of the cGGM is

$$H_\Sigma = E(\nabla_{\Lambda, \Theta}^2 l_2(\Lambda, \Theta)) = \begin{bmatrix} \Lambda^{-1} \otimes (\Lambda^{-1} + 2\Lambda^{-1}\Theta^T \Sigma \Theta \Lambda^{-1}) & -2\Lambda^{-1} \otimes \Lambda^{-1}\Theta^T \Sigma \\ -2\Lambda^{-1} \otimes \Sigma \Theta \Lambda^{-1} & 2\Lambda^{-1} \otimes \Sigma \end{bmatrix} = \begin{bmatrix} H_{1,\Sigma} & H_{2,\Sigma} \\ H_{3,\Sigma} & H_{4,\Sigma} \end{bmatrix}$$

where $H_{1,\Sigma} = \Lambda^{-1} \otimes \Lambda^{-1} + \Lambda^{-1} \otimes A_1 \Sigma B_1$, $H_{l,\Sigma} = \Lambda^{-1} \otimes A_l \Sigma B_l$, for $l = 2, 3, 4$. Then, $|||H_{l,\Sigma} - H_l|||_2 = |||\Lambda^{-1} \otimes A_l(\Sigma - S_{xx})B_l|||_2 \leq |||\Lambda^{-1}|||_2 |||A_l|||_2 |||\Sigma - S_{xx}|||_2 |||B_l|||_2 \leq C_{l,\Gamma} |||\Sigma - S_{xx}|||_2$, where $C_{l,\Gamma}$ is some constant only depending on Λ, Θ , for $l = 1, 2, 3, 4$. With the same permutation matrix P in (1), we

have $H_{SS,\Sigma} = P_1 H_{1,\Sigma} P_1^T + P_2 H_{3,\Sigma} P_1^T + P_1 H_{2,\Sigma} P_2^T + P_2 H_{4,\Sigma} P_2^T$, and $H_{\bar{S}S,\Sigma} = P_3 H_{1,\Sigma} P_1^T + P_4 H_{3,\Sigma} P_1^T + P_3 H_{2,\Sigma} P_2^T + P_4 H_{4,\Sigma} P_2^T$. Then,

$$\begin{aligned}
& |||H_{\Sigma,SS} - H_{SS}|||_2 \\
& \leq |||P_1|||_2 (|||H_{1,\Sigma} - H_1|||_2 |||P_1^T|||_2 + |||H_{2,\Sigma} - H_2|||_2 |||P_2^T|||_2) \\
& \quad + |||P_2|||_2 (|||H_{3,\Sigma} - H_3|||_2 |||P_1^T|||_2 + |||H_{4,\Sigma} - H_4|||_2 |||P_2^T|||_2) \\
& \leq |||P_1|||_2 (C_{1,\Gamma} |||\Sigma - S_{xx}|||_2 |||P_1^T|||_2 + C_{2,\Gamma} |||\Sigma - S_{xx}|||_2 |||P_2^T|||_2) \\
& \quad + |||P_2|||_2 (C_{3,\Gamma} |||\Sigma - S_{xx}|||_2 |||P_1^T|||_2 + C_{4,\Gamma} |||\Sigma - S_{xx}|||_2 |||P_2^T|||_2) \\
& \leq C_{SS} |||\Sigma - S_{xx}|||_2,
\end{aligned}$$

where $C_{SS} = |||P_1|||_2 (C_{1,\Gamma} |||P_1^T|||_2 + C_{2,\Gamma} |||P_2^T|||_2) + |||P_2|||_2 (C_{3,\Gamma} |||P_1^T|||_2 + C_{4,\Gamma} |||P_2^T|||_2)$. Similarly, we have $|||H_{\Sigma,\bar{S}S} - H_{\bar{S}S}|||_2 \leq C_{\bar{S}S} |||\Sigma - S_{xx}|||_2$, where $C_{\bar{S}S} = |||P_3|||_2 (C_{1,\Gamma} |||P_1^T|||_2 + C_{2,\Gamma} |||P_2^T|||_2) + |||P_4|||_2 (C_{3,\Gamma} |||P_1^T|||_2 + C_{4,\Gamma} |||P_2^T|||_2)$. Now for the proof of Lemma 1, Theorem 1.2 in [?] showed assuming the conditions in Lemma 1 hold, for any $\epsilon > 0$, there exists some $C_* > 0$, with probability at least $1 - \epsilon$, Σ can be approximated by $S_{xx} = X^T X/n$ as

$$|||\Sigma - S_{xx}|||_2 \leq C_* (\log \log d)^2 \left(\frac{d}{n}\right)^{\frac{1}{2} - \frac{2}{q}}. \quad (3)$$

Note that $|||S_{xx}|||_2 \leq |||S_{xx} - \Sigma|||_2 + |||\Sigma|||_2$. Thus, when $n > \left(C_*(\log \log d)^2 d^{1/2-2/q}\right)^{-1/(2/q-1/2)}$, $|||S_{xx} - \Sigma|||_2 < 1$, $|||S_{xx}|||_2 \leq 1 + |||\Sigma|||_2 < \infty$ with probability at least $1 - \epsilon$. Then, $|||H_l|||_2 \leq |||\Lambda^{-1}|||_2 |||A_l|||_2 |||S_{xx}|||_2 |||B_l|||_2 + I_{\{l=1\}} |||\Lambda^{-1} \otimes \Lambda^{-1}|||_2 \leq C_{l,T} (1 + |||\Sigma|||_2) + I_{\{l=1\}} |||\Lambda^{-1} \otimes \Lambda^{-1}|||_2 = C_{l,S} < \infty$ with probability at least $1 - \epsilon$, where $C_{l,T} = |||\Lambda^{-1}|||_2 |||A_l|||_2 |||B_l|||_2$. Hence, there exists a constant C_S such that $|||H_{\bar{S}S}|||_2 \leq |||P_3|||_2 (C_{1,S} |||P_1^T|||_2 + C_{2,S} |||P_2^T|||_2) + |||P_4|||_2 (C_{3,S} |||P_1^T|||_2 + C_{4,S} |||P_2^T|||_2) = C_S < \infty$ with probability at least $1 - \epsilon$. Let $C_\Sigma = |||H_{\Sigma,SS}^{-1}|||_2$, and

$$C_H = C_{\bar{S}S} C_\Sigma + C_S C_P C_\Sigma C_{SS}. \quad (4)$$

Then, when $n > (\max\{C_*, C_{**}\} (\log \log d)^2 d^{1/2-2/q})^{-1/(2/q-1/2)}$, $|||\Sigma - S_{xx}|||_2 <$

$\min\{1, C_\alpha/C_H\}$ with probability at least $1 - \epsilon$, and we have

$$\begin{aligned}
& |||H_{\Sigma, \bar{S}S}H_{\Sigma, SS}^{-1} - H_{\bar{S}S}H_{SS}^{-1}|||_2 \\
& = |||H_{\Sigma, \bar{S}S}H_{\Sigma, SS}^{-1} - H_{\bar{S}S}H_{\Sigma, SS}^{-1} + H_{\bar{S}S}H_{\Sigma, SS}^{-1} - H_{\bar{S}S}H_{SS}^{-1}|||_2 \\
& \leq \left(|||H_{\Sigma, \bar{S}S}H_{\Sigma, SS}^{-1} - H_{\bar{S}S}H_{\Sigma, SS}^{-1}|||_2 + |||H_{\bar{S}S}H_{\Sigma, SS}^{-1} - H_{\bar{S}S}H_{SS}^{-1}|||_2 \right) \\
& \leq \left(C_{\bar{S}S} |||\Sigma - S_{xx}|||_2 C_\Sigma + |||H_{\bar{S}S}|||_2 |||H_{\Sigma, SS}^{-1}(H_{SS} - H_{\Sigma, SS})H_{SS}^{-1}|||_2 \right) \\
& \leq \left(C_{\bar{S}S} |||\Sigma - S_{xx}|||_2 C_\Sigma + C_S |||H_{\Sigma, SS}^{-1}|||_2 |||H_{SS} - H_{\Sigma, SS}|||_2 |||H_{SS}^{-1}|||_2 \right) \\
& \leq C_{\bar{S}S} |||\Sigma - S_{xx}|||_2 C_\Sigma + C_S C_P C_\Sigma |||H_{SS} - H_{\Sigma, SS}|||_2 \\
& \leq C_{\bar{S}S} |||\Sigma - S_{xx}|||_2 C_\Sigma + C_S C_P C_\Sigma C_{SS} |||\Sigma - S_{xx}|||_2 \\
& = C_H |||\Sigma - S_{xx}|||_2 \\
& < C_\alpha
\end{aligned}$$

with probability at least $1 - \epsilon$. Since $|||\cdot|||_\infty \leq |||\cdot|||_2$, we have $|||H_{\Sigma, \bar{S}S}H_{\Sigma, SS}^{-1} - H_{\bar{S}S}H_{SS}^{-1}|||_\infty < C_\alpha$, with probability at least $1 - \epsilon$. $\square$